Rain and snow

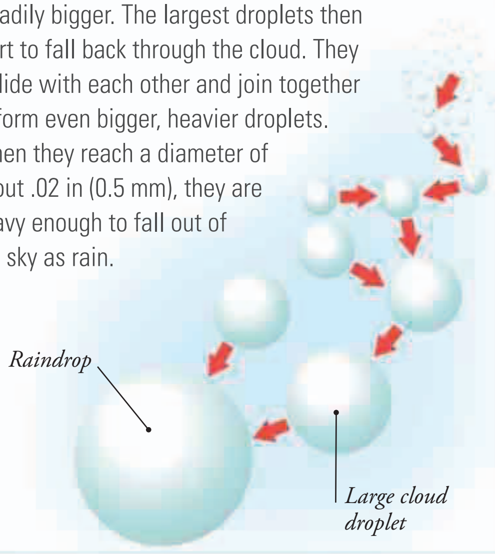
Tiny water droplets in clouds can join together to form bigger drops that are heavy enough to fall toward the ground. Ice crystals in high-altitude clouds may do the same, joining together into snowflakes. Before reaching the ground, small water drops may evaporate and snowflakes may melt. Where they do reach ground level, they fall as rain and snow.

THREATENING CLOUDS

You know that it's likely to rain when you see dark clouds approaching. They're dark because they're full of big water droplets that block out the light that makes other clouds look white. You may even be able to see rain falling out of the clouds, forming a gray curtain that hangs down from the base of the cloud.

RAINDROPS

As cloudy air rises, the water droplets grow steadily bigger. The largest droplets then start to fall back through the cloud. They collide with each other and join together to form even bigger, heavier droplets. When they reach a diameter of about .02 in (0.5 mm), they are heavy enough to fall out of the sky as rain.



RAIN AND SHOWERS

The sheets of shallow cloud that form in cyclones produce light, persistent rain. The clouds are not deep enough to allow big raindrops to form, and the drops that do fall often evaporate before they reach the ground. Deep convection clouds cause intense, localized showers. The depth of the cloud allows the raindrops to grow much bigger, and they often fall in dramatic cloudbursts.

▼ RAINY MOOR
On high moorland, sheets of shallow cloud can produce a widespread drizzle that lasts all day.